

JPE Replication Report

PUBLISHED
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1 Paper Identification

ID	20251543
Author	Giorcelli
Title	Copyright and Creativity: Evidence from Italian Opera in the Napoleonic Age
Iteration	1

Some useful information up front:

- [JPE Data and Code Availability Policy](#)
- [Step by step guidance from Data Editor for Authors](#)
- [Template README](#)

2 Summary

In assessing compliance with our [Data and Code Availability Policy](#), our reproducibility team has identified the following issues, which we ask you to address:

1. All substantive points of GM20 are verified with the supplied raw data. The main issue with the original package is a wrong variable `season_count`, which was indeed collinear with another variable. Fixing this does not change the results of the paper.
2. The remaining allegations are successfully refuted with the raw data at hand. GM20 point out significant econometric errors in the comment (more parameters than observations in a linear regression framework, inappropriate clustering, etc)
3. We could not reproduce table 6 for the reasons outlined in point 2. This table aims to replicate results in the comment MT25.
4. The replication package is not conforming to our current standards in terms of documentation and structure. We stand by to move forward with suggestions for improvements if this case moves ahead.

3 General

- ☐ Data Availability and Provenance Statements
 - ☐ Statement about Rights - Part 1: Right to use the data (e.g. *did the authors lawfully obtain the data*)
 - ☐ Statement about Rights - Part 2: Right to publish the data (e.g. *are human subjects involved and did permission been granted to publish their data?*)
 - ☐ License for Data (optional, but recommended)
 - ☐ Details on each Data Source
- ☒ Dataset list
- ☐ Computational requirements
 - ☒ Software Requirements
 - ☐ Controlled Randomness (as necessary)
 - ☐ Memory, Runtime, and Storage Requirements
- ☒ Description of programs/code
 - ☐ License for Code (Optional, but recommended)
- ☒ List of tables and programs
- ☐ References (Optional, but recommended)

4 High-level Description of Research Project

This research project...

- ☒ uses observational data.
- ☐ uses *confidential* observational data.
- ☐ uses data generated via experiment or trial by the authors (in field or lab).
 - ☐ For lab experiments: The code to implement the experimental protocol (z-Tree, o-Tree etc) is included in the package.
 - ☐ A PDF copy of IRB approval is contained in the package.
 - ☐ A PDF document outlining the design of the experiment, including information on the selection and eligibility of participants is included.
 - ☐ A PDF copy of the instructions given to participants in both original language and an English translation is included.
- ☐ uses data generated via survey by the authors (in field or online)

- ☐ The programs used to administer the survey (e.g. qualtrics survey file) are included.
- ☐ A PDF copy of IRB approval is contained in the package.
- ☐ A PDF of the original questionnaire(s) and information on the selection and eligibility of participants is included.
- ☐ is a simulation study: the data are generated by an algorithm.
- ☐ is a numerical illustration: no data are used but code is used to produce illustrative output (tables, figures)

5 Data description

5.1 Data Sources

file/name	public	private	DOI/URL	Access described	cited readme	cited paper
operas_raw_data.dta	yes	no	-	no	yes	no
theater_raw_data.dta	yes	no	-	no	yes	no
additional_operas.dta	yes	no	-	no	yes	no
composer_roster.dta	yes	no	-	no	yes	no

6 Requirements

{REPLICATOR}: Check that these requirements are met.

- ☒ README is in TXT, MD, or PDF format
- ☒ README is accessible at the root of the package (not inside a folder)
- ☐ Title conforms to guidance (starts with "Data and Code for: Title of Paper" or "Code for: Title of Paper", is properly capitalized)
- ☐ Authors (with affiliations) are listed in the same order as on the paper

{REQUIRED} Please review the title of the deposit as per our guidelines.

{REQUIRED} Please review authors and affiliations on the deposit. In general, they are the same, and in the same order, as for the manuscript; however, authors can deviate from that order.

6.1 Stated Requirements

- ☐ No requirements specified
- ☒ Software Requirements specified as follows:
 - Stata/MP Version 17
- ☐ Computational Requirements specified as follows:
 - Cluster size, etc.
- ☐ Time Requirements specified as follows:
 - Length of necessary computation (hours, weeks, etc.)
- ☐ Requirements are complete.

For missing requirements, see the list of required changes in [Section 9](#)

7 Analyzing Data and Code Files

7.1 Potential Personal Identifiable Information (PII)

⚠ We found the following instances of potentially personally identifying information. This may be completely legitimate but might be worth checking. As a reminder, privacy legislation in many countries (e.g. GDPR in EU) prohibits the dissemination of personal identifiable information without prior (and documented) consent of individuals. If indeed you want to publish such information with your replication package, you should probably have obtained IRB approval for this - please check!

Summary: - Data files with PII indicators: 5 - Variables flagged in data: 16 - Code files with PII references: 9 - PII references in code: 226

7.1.1 Summary of Flagged Files

File Type	File	Variables/References	PII Categories
Data	additional_operas.dta	1	name
Data	city_level_operas_1781_1820.dta	6	city
Data	composers_return_year.dta	1	name
Data	operas_raw_data.dta	3	city, name, son
Data	theater_raw_data.dta	5	city
Code	GM20_city_level_analysis.do	45	city, loc
Code	GM20_composer_level_analysis.do	24	name, loc
Code	GM20_state_level_analysis.do	112	city, son, loc
Code	Replication_PreTrends.do	2	loc
Code	Replication_Table3_GM20.do	12	city
Code	Replication_Table4_MT25.do	23	city, son
Code	Replication_Table5_MT25.do	2	son
Code	Replication_Table7_MT25.do	2	loc
Code	Replication_TableB1_MT25.do	4	son

See [Appendix](#) for detailed listing of all flagged instances.

7.1.2 File Paths Report

Generated on 2026-06-03T10:18:57.328

Warning: Our search on file path types is imperfect and incurs both type 1 and type 2 errors. We aim to strike a reasonable balance between both. The below table is therefore only indicative. Detailed listings can be found in the appendix to this report.

In this table we analyse all files which contain source code (not data and documentation), and we report the grand total of each type of filepath we encountered.

Please check and replace any windows filepaths with unix compliant paths, insofar as this is possible in your setup. (Notice that STATA allows / to be a filepath separator on a windows platform, hence this is a requirement for all STATA applications.)

Files Analyzed	Windows Paths	Unix Paths	Mixed Paths	Drive Letters (C:\ etc)
10	0	1	0	0

7.1.3 Duplicate Files Report

We did not find any duplicate files.

7.1.4 Zero Size Files Report

We did not find any zero sized files.

7.1.5 Large Files Report

We did not find any files larger than 100MB.

7.2 Code description

There are **10 provided Stata do files** in total. Three are **data preparation + analysis files** (GM20_state_level_analysis.do, GM20_city_level_analysis.do, GM20_composer_level_analysis.do) that build datasets from raw data and produce the main GM20 outputs. The remaining seven (Replication_Table3_GM20.do, Replication_Table4_MT25.do through Replication_TableB1_MT25.do, Replication_PreTrends.do) are pure **analysis files** that load intermediate datasets and produce output tables. There is **no master do file** that runs all programs sequentially.

The mapping of outputs to programs is as follows:

Output	Program	Approx. lines
Table 1	GM20_state_level_analysis.do	124–155
Table 3	GM20_state_level_analysis.do	163–192
Table 4	GM20_state_level_analysis.do	200–236
Table 5	GM20_composer_level_analysis.do	81–227
Table 8	GM20_city_level_analysis.do	89–138
Figure 2	GM20_state_level_analysis.do	244–250
Figure 5	GM20_composer_level_analysis.do	244–248
Table A3	GM20_state_level_analysis.do	256–389
Table A7	GM20_state_level_analysis.do	393–422
Table A8	GM20_state_level_analysis.do	430–542
Table A9	GM20_state_level_analysis.do	548–603
Table A10	GM20_state_level_analysis.do	610–653
Response Table 1 (GM20_Table3)	Replication_Table3_GM20.do	31–150
Response Table 2 (tableA10)	GM20_state_level_analysis.do	610–653
Response Table 3 (MT25_Table4)	Replication_Table4_MT25.do	75–121
Response Table 4 (MT25_Table5)	Replication_Table5_MT25.do	30–52
Response Table 5 (MT25_Table6)	Replication_Table6_MT25.do	32–38
Response Table 6 (MT25_Table7)	Replication_Table7_MT25.do	34–65
Response Table 7 (Pretrends)	Replication_PreTrends.do	32–59
Response Table 8 (MT25_TableB1)	Replication_TableB1_MT25.do	33–51
Response Table 9 (MT25_TableB1_Excluding_Modena_Parma)	Replication_TableB1_MT25.do	66–89

cloc		github.com/AIDanial/cloc v 2.02 T=0.04 s (714.4 files/s, 47528.3 lines/s)		
Language	files	blank	comment	code
Stata	10	414	0	1223
Text	20	39	0	320
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SUM:	30	453	0	1543

☐ The replication package contains a “main” or “master” file(s) which calls all other auxiliary programs.

{SUGGESTED} We strongly advise the use of a single (or a small number of) main control file(s) to automatically reproduce all figures and tables in the paper, without manual interaction.

NOTE: In-text numbers that reference numbers in tables do not need to be listed. Only in-text numbers that correspond to no table or figure need to be listed.

7.3 Computing Environment of the Replicator

- Nuvolos cloud environment, Linux 6.12.62, 503 GB of memory, (AMD EPYC 9354P 32-Core Processor, 64 logical CPUs / 32 cores)
- Stata/MP 17.0 (4CPU, 16G RAM)

8 Replication

All code ran smoothly, with the small exception that some STATA packages had to be installed. These are detailed in @missing-requirements.

9 Findings

Summarizing the Findings:

README conventions:

- Title does not conform to JPE guidance (should start with "Data and Code for: ...")
- Authors and affiliations not verified against manuscript order
- No statement on rights to use the data
- No statement on rights to publish the data
- No DOI or URL provided for any of the four data sources
- Access procedure not described for any of the four data sources
- Data sources not cited in the paper
- No references section
- `composers_return_year.dta` is used in Figure 5 but not listed in README
- No computational requirements stated (memory, runtime, storage)
- No license for code or data provided

Missing requirements:

- Stata packages not listed in README; outreg2 had to be installed manually during replication.
- No master do-file to run all programs sequentially

Reproduction:

- All GM20 original tables and figures reproduced successfully, though no manuscript was available for direct comparison
- Response Tables 1–5 and 7–9 are partially reproduced; differences are minor (missing Panel A entries, trivial rounding, cosmetic presentation) with the notable exception of a recurring Poisson discrepancy in cols 5 of Response Tables 8 and 9
- Response Table 2 columns 1–3 cannot be reproduced as originally reported, expectedly so given the authors corrected the `season_count` variable in the updated package
- Response Table 6 is not reproduced, with substantial differences in sample size (77 vs 231) and several coefficients

Substantiation of claims:

- The authors' central claims are substantiated across all response tables where reproduction was successful; the treatment effect of copyright on opera production is consistently positive and significant across aggregation levels, clustering approaches, pre-trend specifications, and sample restrictions
- Response Table 6 is the only table where confidence in the substantiation is limited due to unresolved discrepancies

9.1 Missing Requirements

The following STATA packages had to be manually installed:

- outreg2
- ☐ Software Requirements
 - ☒ Stata
 - ☒ Version 17
 - Packages go here
 - ☐ Matlab
 - ☐ Version
 - ☐ R
 - ☐ Version
 - R packages go here
 - ☐ Python
 - ☐ Version
 - Python package go here
 - ☐ REPLACE ME WITH OTHER
- ☐ Computational Requirements specified as follows:
 - Cluster size, disk size, memory size, etc.
- ☐ Time Requirements
 - Length of necessary computation (hours, weeks, etc.)

[REQUIRED] Please amend README to contain complete requirements.

You can copy the section above, amended if necessary.

9.2 Data Preparation Code

All code ran without error, with the small exception that some STATA packages had to be installed. These are detailed in @missing-requirements.

9.3 Tables and Figures

Output	Program	Approx. lines	Reproduced?	Comments	Substantiates claim?
Table 1	GM20_state_level_analysis.do	124–155	Yes - though no manuscript to compare to	-	NA
Table 3	GM20_state_level_analysis.do	163–192	Yes - though no manuscript to compare to	-	NA
Table 4	GM20_state_level_analysis.do	200–236	Yes - though no manuscript to compare to	-	NA
Table 5	GM20_composer_level_analysis.do	81–227	Yes - though no manuscript to compare to	-	NA
Table 8	GM20_city_level_analysis.do	89–138	Yes - though no manuscript to compare to	-	NA
Figure 2	GM20_state_level_analysis.do	244–250	Yes - though no manuscript to compare to	-	NA
Figure 5	GM20_composer_level_analysis.do	244–248	Yes - though no manuscript to compare to	composers_return_year.dta not listed in README	NA
Table A3	GM20_state_level_analysis.do	256–389	Yes - though no manuscript to compare to	-	NA
Table A7	GM20_state_level_analysis.do	393–422	Yes - though no manuscript to compare to	-	NA
Table A8	GM20_state_level_analysis.do	430–542	Yes - though no manuscript to compare to	-	NA

Output	Program	Approx. lines	Reproduced?	Comments	Substantiates claim?
Table A9	GM20_state_level_analysis.do	548–603	Yes - though no manuscript to compare to	-	NA
Table A10	GM20_state_level_analysis.do	610–653	Yes - though no manuscript to compare to	-	NA
Response Table 1 (GM20_Table3)	Replication_Table3_GM20.do	31–150	Yes - partly	Coefficients and SEs match. Poisson column 5 missing. R-squared values differ for columns 2–4. Pre-1801 mean row missing. Typos — “City-Yearr” and “Composer-Yearr” in columns 8 and 12. Variable name — “copyright_post1801” vs “Lombardy & Venetia * post”. See screenshots below.	Table 1: Yes: the treatment effect is positive and statistically significant across all three aggregation levels (state, city, composer), confirming that GM20’s main result is robust to alternative dataset collapse.
Response Table 2 (tableA10)	GM20_state_level_analysis.do	610–653	Yes - partly	Columns 1–3 not replicated. Different coefs, R-squared, pre-1801 means.	Yes: columns 4–6 (the corrected results the authors stand behind) match exactly; divergence in columns 1–3 is expected as the package has been updated with the corrected season_count variable.
Response Table 3 (MT25_Table4)	Replication_Table4_MT25.do	75–121	Yes - partly	Panel A (MT25’s original results) is absent; two trivial rounding differences in cols 4 and 8 (1.726 vs 1.725 and 1.468 vs 1.467).	Yes: treatment effect remains positive and significant across all exclusion robustness checks, contradicting MT25’s assertion that results are driven by Milan and Venice.

Output	Program	Approx. lines	Reproduced?	Comments	Substantiates claim?
Response Table 4 (MT25_Table5)	Replication_Table5_MT25.do	30–52	Yes - partly	Panel A (MT25's original results) is absent; col 5 Poisson loses significance (1.287 with no stars vs 1.287* in original), with a trivial standard error difference (0.783 vs 0.782).	Yes: the authors' key point is that at conventional significance levels cols 1–4 remain significant even with state-level clustering, and this holds in the replication; the marginal significance loss in col 5 does not affect this conclusion.
Response Table 5 (MT25_Table6)	Replication_Table6_MT25.do	32–38	Yes - partly	Panel A (MT25's unreplicable results) is absent, which is expected given the letter explicitly states MT25's specification is econometrically impossible; Panel B matches exactly.	Yes: the correctly implemented Bertrand et al. (2004) procedure yields significant results at 5% and 10% respectively, supporting the authors' argument that GM20's findings hold even under this alternative approach to serial correlation.
Response Table 6 (MT25_Table7)	Replication_Table7_MT25.do	34–65	No	Coefficients and standard errors differ substantially across multiple columns — col 1 (1.462 matches but SE differs: 1.335 vs 2.313), col 2 (0.150 matches but SE differs: 0.152 vs 0.263), col 3 (0.270 vs 0.037, with significance appearing in replication but not original), col 5 (0.238 gains significance in replication); observations differ drastically (77 vs 231 throughout), suggesting a different sample restriction has been applied.	Partially: the letter acknowledges MT25's results are not replicable and notes the authors' own Panel B yields a col 1 coefficient of 1.462 — which matches here — but the dramatically different sample size (77 vs 231) and several differing coefficients

Output	Program	Approx. lines	Reproduced?	Comments	Substantiates claim?
					mean this replication cannot be directly compared to the letter's Table 6 with confidence.
Response Table 7 (Pretrends)	Replication_PreTrends.do	32–59	Yes - partly	All four main coefficients and standard errors match exactly; the replication shows the underlying state-specific pre-trend coefficients in cols 3 and 4 which are not reported in the original, but this is additional detail rather than a discrepancy.	Yes: all four treatment effect estimates remain virtually identical regardless of pre-trend specification, confirming the authors' argument that MT25's critique of the pre-trend construction has no material impact on GM20's results.
Response Table 8 (MT25_TableB1)	Replication_TableB1_MT25.do	33–51	Yes - partly	Cols 1–4 coefficients and standard errors match exactly; col 5 Poisson coefficient differs (1.221 vs 0.639) and standard error differs (0.306 vs 0.163); R-squared in col 2 differs trivially (0.675 vs 0.677).	Yes: the key OLS results in cols 1–4 match exactly, confirming the authors' claim that setting the 46 disputed state-year pairs to zero leaves results positive and significant; the Poisson discrepancy in col 5 does not affect this conclusion.
Response Table 9 (MT25_TableB1_Excluding_Modena_Parma)	Replication_TableB1_MT25.do	66–89	Yes - partly	Cols 1–4 coefficients and standard errors match exactly; col 5 Poisson coefficient differs (1.223 vs 0.510) and standard error differs (0.383 vs 0.162), mirroring the same Poisson discrepancy seen in Table 8.	Yes: the key OLS results in cols 1–4 match exactly, confirming the authors' claim that results remain positive and significant after excluding Modena and

Output	Program	Approx. lines	Reproduced?	Comments	Substantiates claim?
					Parma; the recurring Poisson discrepancy in col 5 warrants attention but does not undermine the main conclusion.

9.4 In-Text Numbers

- ☒ In-text numbers not verified.
- ☐ There are no in-text numbers, or all in-text numbers stem from tables and figures.
- ☐ There are in-text numbers, but they are not identified in the code.

10 Classification

- ☐ full reproduction
- ☐ full reproduction with minor issues
- ☒ partial reproduction (see above)
- ☐ not able to reproduce most or all of the results (reasons see above)

10.1 Reason for incomplete reproducibility

- ☐ None.
- ☒ **Discrepancy in output** (either figures or numbers in tables or text differ)
- ☐ **Bugs in code** that were fixable by the replicator (but should be fixed in the final deposit)
- ☐ **Code missing**, in particular if it prevented the replicator from completing the reproducibility check

☐ **Data preparation code missing** should be checked if the code missing seems to be data preparation code
- ☐ **Code not functional** is more severe than a simple bug: it prevented the replicator from completing the reproducibility check
- ☐ **Software not available to replicator** may happen for a variety of reasons, but in particular (a) when the software is commercial, and the replicator does not have access to a licensed copy, or (b) the software is open-source, but a specific version required to conduct the reproducibility check is not available.
- ☐ **Insufficient time available to replicator** is applicable when (a) running the code would take weeks or more (b) running the code might take less time if sufficient compute resources were to be brought to bear, but no such resources can be accessed in a timely fashion (c) the replication package is very complex, and following all (manual and scripted) steps would take too long.
- ☐ **Data missing** is marked when data *should* be available, but was erroneously not provided, or is not accessible via the procedures described in the replication package
- ☐ **Data not available** is marked when data requires additional access steps, for instance purchase or application procedure.
- ☐ **Missing README** is marked if there is no README to guide the replicator, or the README is not in compliance with JPE requirements

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Summary: - Data files with PII indicators: 5 - Variables flagged in data: 16 - Code files with PII references: 9 - PII references in code: 226

10.2.1 Summary of Flagged Files

File Type	File	Variables/References	PII Categories
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Data	city_level_operas_1781_1820.dta	6	city

File Type	File	Variables/References	PII Categories
Data	composers_return_year.dta	1	name
Data	operas_raw_data.dta	3	city, name, son
Data	theater_raw_data.dta	5	city
Code	GM20_city_level_analysis.do	45	city, loc
Code	GM20_composer_level_analysis.do	24	name, loc
Code	GM20_state_level_analysis.do	112	city, son, loc
Code	Replication_PreTrends.do	2	loc
Code	Replication_Table3_GM20.do	12	city
Code	Replication_Table4_MT25.do	23	city, son
Code	Replication_Table5_MT25.do	2	son
Code	Replication_Table7_MT25.do	2	loc
Code	Replication_TableB1_MT25.do	4	son

See [Appendix](#) for detailed listing of all flagged instances.